

## MSCF Probability – Final Exam – August 19, 2019

Name \_\_\_\_\_

Email \_\_\_\_\_

### Instructions:

- You should complete all questions on this exam by writing out the solutions on paper that will be provided to you. Anything written on the exam paper itself will **not** be graded.
- You will have two hours to complete this exam. No matter when you start the exam, you have to stop at 11:00 AM.
- Notes (except described below), books and calculators/laptops/phones are not allowed.
- A table of common distributions, with pdfs/pmfs, and other information, is included at the end of this exam.
- Final answers can involve any of the following notation:

$$\binom{n}{k}, n^k, \exp(k), n!, \Gamma(x), \Phi(x),$$

where  $\Phi(x)$  is the CDF of the standard normal distribution. Except for cases in which it is explicitly stated, final answers should **not** include integrals.

- You may use 2 handwritten sides of a 8.5” by 11” page (1 double-sided or 2 single-sided) of notes.
- Questions are not all worth the same number of points. The numbers in **bold** after each question indicate the number of points. Also, you may find some easier than others. You should prioritize your work on particular questions accordingly.

**Note:** The total number of points is **58**.

- You **must show all of your work**, except for on the multiple choice questions. A correct answer without explanation will receive only minimal or no partial credit. Note that the “easier” a question may seem, the more burden you have to show that you understand how you are reaching the answer; again, **be clear and complete in your work**.

**PLEASE CHECK THAT YOU HAVE PAGES 1 – 6,  
THE DISTRIBUTION TABLE, AND THE MATH FACTS**

## Grading Guidelines:

The following are reasons why you may lose points:

1. Failure to mention when an independence assumption is utilized.
2. Failure to mention when the assumption that random variables are uncorrelated is utilized.
3. Failure to specify clearly the support of a distribution.

You do **not** need to cite the names of rules and definitions when you use them, such as

1. The Rule for the Lazy Statistician
2. The Iterated Expectation Rule
3. The Rule for Conditional Probability
4. The Addition Rule
5. The Chain Rule
6. ...

But, in general, any work that does not make it clear how the answer was found is subject to losing points. In other words, even if you do not justify every step, there should be sufficient number of steps to make it clear you know how to do the derivation. Lack of clarity can also result from messy writing: write neatly.

1.  $X$  has the Exponential( $\lambda$ ) distribution, i.e.  $E(X) = 1/\lambda$ . What is  $E(X^2)$ ? **(5)**  
 $2/\lambda^2$

2. Which of the following are always true regarding  $P(A|B)$ ? **(4)**  
**(CHOOSE AS MANY AS APPROPRIATE)**

- (a)  $P(A|B)$  is only defined if  $P(B) > 0$ .
- (b)  $P(A|B) = P(A \cap B)/P(B)$  if  $P(B) > 0$ . **YES**
- (c)  $P(A|B) \geq P(A)$  for any events  $A, B$ .
- (d)  $P(A|B) = P(A)$  when  $A, B$  are independent events. **YES**

3. A fair coin is tossed twice. Define

$$\begin{aligned}A_1 &= \{\text{First toss is "heads"}\} \\A_2 &= \{\text{Second toss is "heads"}\} \\B &= \{\text{the two tosses have different outcomes}\}\end{aligned}$$

In other words,  $B$  is the event that exactly one of the two tosses is “heads.”

What can be said about these events? **(4)**

**(CHOOSE AS MANY AS APPROPRIATE)**

- (a) Events  $A_1$  and  $A_2$  are independent. **YES**
  - (b) Events  $A_1$  and  $A_2$  are independent conditional on  $B$ .
  - (c) Events  $A_1$  and  $B$  are independent. **YES**
4. Suppose that  $X$  is a random variable and  $Y = X + b$ , where  $b$  is a constant. What is the relationship between the moment generating function (MGF) for  $X$  and the MGF for  $Y$ ? **(CHOOSE ONE) (4)**

- (a)  $M_Y(t) = e^{tb}M_X(t)$  **CORRECT**
- (b)  $M_Y(t) = M_X(tb)$
- (c)  $M_Y(t) = M_X(t)$
- (d)  $M_Y(t) = bM_X(t)$

5. A factory produces  $X_n$  gadgets on day  $n$ . It can be reasonably assumed that  $X_1, X_2, \dots$  are independent and identically distributed random variables, each with mean 5 and variance 9. Approximate the probability that the number of gadgets produced in 100 days is fewer than 440. **(5)**

Use CLT. The desired probability is approximately  $\Phi(-2) = 0.022$ .

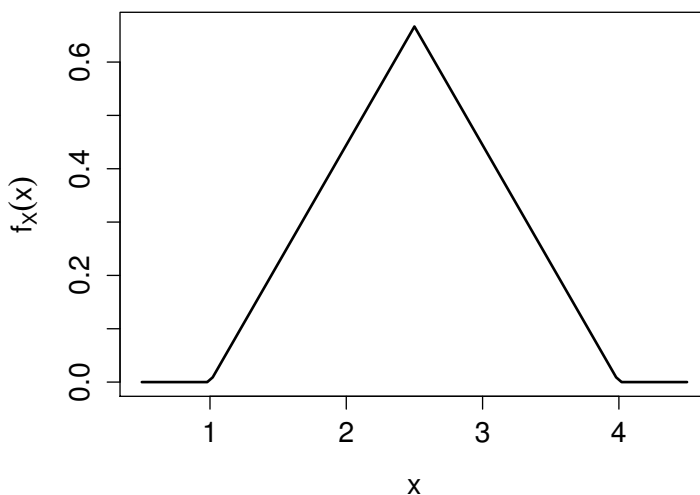
6. Suppose a stick of length one is broken by choosing a point uniformly on 0 to 1. This creates two pieces, call them the “left” and “right” pieces. The “right” piece is discarded, and the “left” piece is then broken again at a position that is uniformly distributed along its length. Again, the “right” piece is discarded. Let  $X$  denote the length of the remaining piece. What is the expected value of  $X$ ? (5)

Let  $Y$  denote the position of the first break. Then  $E(X) = E(E(X|Y)) = E(0.5Y) = 0.5E(Y) = 0.25$ .

7. Suppose  $a < b$ . A random variable  $X$  is said to have the *triangular( $a, b$ ) distribution* if it has probability density function (pdf) of

$$f_X(x) = \begin{cases} \frac{2}{b-a} \left( 1 - \frac{2}{b-a} \left| x - \frac{a+b}{2} \right| \right), & a < x < b \\ 0, & \text{otherwise} \end{cases}$$

The figure below shows this pdf for the case where  $a = 1$  and  $b = 4$ .



Some other useful properties of this distribution: If  $X$  is triangular( $a, b$ ), then  $E(X) = (a + b)/2$  and the moment generating function is, for  $t \neq 0$ ,

$$M_X(t) = \left( \frac{e^{tb/2} - e^{ta/2}}{t(b-a)/2} \right)^2$$

Suppose that  $U_1$  and  $U_2$  are independent, and each has the Uniform( $a/2, b/2$ ) distribution. Show that  $U_1 + U_2$  has the triangular( $a, b$ ) distribution. (6)

Follows directly from property of MGFs.

8. Suppose that  $U_1, U_2, U_3$  are i.i.d. random variables whose values are determined by the flips of a fair coin. Hence, each has probability mass function (pmf) as follows:

$$f_{U_i}(k) = \begin{cases} 0.5, & \text{if } k = 3 \text{ or } k = 1/3 \\ 0, & \text{otherwise} \end{cases}$$

We can think of each of the  $U_i$  as being defined on the sample space

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

Now, define  $S_n$  to be the product of the first  $n$  of the  $U_i$ . In other words,  $S_1 = U_1$ ,  $S_2 = U_1U_2$ , and  $S_3 = U_1U_2U_3$ .

Define  $\mathcal{F}_n = \sigma(U_1, U_2, \dots, U_n)$  and  $\mathcal{G}_n = \sigma(S_n)$ .

**Comment:** Note the similarity between this setup and the binomial pricing models we considered during lecture.

- (a) Which of the following is definitely true for any choice of  $n$ ? **(CHOOSE ONE) (2)**

- i.  $\mathcal{G}_n \subseteq \mathcal{F}_n$  **CORRECT**
- ii.  $\mathcal{F}_n \subseteq \mathcal{G}_n$
- iii. Both (i) and (ii) above are true, i.e.,  $\mathcal{F}_n = \mathcal{G}_n$ .

- (b) What is  $E(S_3|\mathcal{G}_2)$ ? **(3)**

**Comment:** Recall that a conditional expectation of this form is a random variable. Hence, you can give your answer by using the table below. Draw it in your exam book, and fill in the table entries. **Fill in the entire table, but you only need to explain the derivation of one entry.**

| $\omega$ | $E(S_3 \mathcal{G}_2)$ |
|----------|------------------------|
| $HHH$    | 15                     |
| $HHT$    | 15                     |
| $HTH$    | 5/3                    |
| $HTT$    | 5/3                    |
| $THH$    | 5/3                    |
| $THT$    | 5/3                    |
| $TTH$    | 5/27                   |
| $TTT$    | 5/27                   |

In this solution, it has been assumed that “Heads” corresponds to multiplying by 3 ( $U_i = 3$ ). It is equally correct to do it the other way around, with “Tails” indicating  $U_i = 3$ . The entries of the table would shift correspondingly.

9. A professor uses an automated cheating detection system. Of course, the system is not perfect. In fact, 10% of papers which are plagiarized are missed by the system, and 5% of papers which are not plagiarized are falsely reported as being plagiarized. Assume that 5% of all submitted papers will be plagiarized. If a randomly chosen paper is reported as being plagiarized, what is the probability that it actually was plagiarized? **(7)**

**Note:** A paper is plagiarized if it is copied from another source.

Use Bayes' Theorem. The desired probability is 18/37.

10. Suppose that  $(X, Y)$  has joint density given by

$$f_{(X,Y)}(x, y) = \begin{cases} (9 - 6x - 6y + 4xy) / 4, & 0 \leq x < 1, 0 \leq y < 1 \\ 0, & \text{otherwise} \end{cases}$$

What is the marginal density for  $X$ ? **(7)**

$$f_X(x) = \begin{cases} 3/2 - x & 0 \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

11. Let  $\mathbf{X}$  be multivariate normal:  $\mathbf{X} = [X_1 \ X_2 \ X_3]^T \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ , with

$$\boldsymbol{\mu} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

and

$$\boldsymbol{\Sigma} = \begin{bmatrix} 1 & -0.3 & 0 \\ -0.3 & 1 & 0.2 \\ 0 & 0.2 & 2 \end{bmatrix}.$$

What is the distribution of  $2X_1 + X_2$ ? Fully specify the distribution, including the mean and variance. **(6)**

Normal with mean 0 and variance 3.8

# Useful Math Facts

1. **Results Regarding  $e$ :** For any number  $a$ ,

$$e^a = \sum_{i=0}^{\infty} \frac{a^i}{i!} \quad \text{and} \quad e^a = \lim_{n \rightarrow \infty} \left(1 + \frac{a}{n}\right)^n$$

2. **Geometric Series:** If  $|r| < 1$  and  $a$  is an integer,

$$\sum_{k=a}^{\infty} r^k = \frac{r^a}{1-r}$$

3. **The Binomial Theorem:** For  $k = 0, 1, 2, \dots$ ,

$$(\lambda_1 + \lambda_2)^k = \sum_{i=0}^k \binom{k}{i} \lambda_1^i \lambda_2^{k-i}$$

4. **The Gamma Function:**

$$\Gamma(t) = \int_0^{\infty} x^{t-1} e^{-x} dx$$

For any  $t$ , it holds that  $\Gamma(t+1) = t\Gamma(t)$ , and for  $n = 1, 2, \dots$ ,  $\Gamma(n) = (n-1)!$ . Also,  $\Gamma(0.5) = \sqrt{\pi}$ .

5. **Definite Integrals:**

$$\int_0^{\infty} e^{-ax} dx = \frac{1}{a}, \quad \int_0^{\infty} x e^{-ax} dx = \frac{1}{a^2}, \quad \int_0^{\infty} x^m e^{-ax} dx = \frac{\Gamma(m+1)}{a^{m+1}}$$

$$\int_0^{\infty} e^{-ax^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{a}}, \quad \int_0^{\infty} x e^{-ax^2} dx = \frac{1}{2a}, \quad \int_0^{\infty} x^m e^{-ax^2} dx = \frac{\Gamma((m+1)/2)}{2a^{(m+1)/2}}$$

$$\int_0^1 t^{x-1} (1-t)^{y-1} dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = \text{The Beta Function}$$

| Name                 | Density/Mass Function                                                                                                      | MGF                                                                                                       | Mean                                        | Variance                                               |
|----------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------------------------------------------------|
| Normal               | $\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right),$<br>$-\infty < x < \infty, \sigma > 0$       | $\exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$                                                         | $\mu$                                       | $\sigma^2$                                             |
| Log-Normal           | $\frac{1}{x\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\log(x)-\mu)^2}{2\sigma^2}\right),$<br>$0 < x < \infty, \sigma > 0$      | —                                                                                                         | $\exp\left(\mu + \frac{\sigma^2}{2}\right)$ | $e^{2(\mu+\sigma^2)} - e^{2\mu+\sigma^2}$              |
| Gamma                | $\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x},$<br>$0 < x < \infty, \alpha, \beta > 0$                    | $\left(\frac{\beta}{\beta-t}\right)^\alpha, \quad t < \beta$                                              | $\frac{\alpha}{\beta}$                      | $\frac{\alpha}{\beta^2}$                               |
| Exponential          | $\beta e^{-\beta x},$<br>$0 < x < \infty, \beta > 0$                                                                       | $\frac{\beta}{\beta-t}, \quad t < \beta$                                                                  | $\frac{1}{\beta}$                           | $\frac{1}{\beta^2}$                                    |
| Beta                 | $\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1},$<br>$0 < x < 1, \alpha, \beta > 0$ | $1 + \sum_{k=1}^{\infty} \left( \prod_{r=0}^{k-1} \frac{\alpha+r}{\alpha+\beta+r} \right) \frac{t^k}{k!}$ | $\frac{\alpha}{\alpha+\beta}$               | $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$ |
| Uniform (continuous) | $\frac{1}{b-a},$<br>$a \leq x \leq b$                                                                                      | $\frac{e^{bt}-e^{at}}{t(b-a)}, \quad t \neq 0$<br>$1, \quad t = 0$                                        | $\frac{b+a}{2}$                             | $\frac{(b-a)^2}{12}$                                   |
| Bernoulli            | $p^x (1-p)^{1-x},$<br>$x = 0, 1, 0 \leq p \leq 1$                                                                          | $pe^t + (1-p)$                                                                                            | $p$                                         | $p(1-p)$                                               |
| Binomial             | $\binom{n}{x} p^x (1-p)^{n-x},$<br>$x = 0, 1, \dots, n, 0 \leq p \leq 1$                                                   | $[pe^t + (1-p)]^n$                                                                                        | $np$                                        | $np(1-p)$                                              |
| Negative Binomial    | $\binom{r+x-1}{x} p^r (1-p)^x,$<br>$x = 0, 1, \dots, r \in \{1, 2, \dots\}, 0 \leq p \leq 1$                               | $\left(\frac{p}{1-(1-p)e^t}\right)^r, \quad t < \log(1/(1-p))$                                            | $\frac{r(1-p)}{p}$                          | $\frac{r(1-p)}{p^2}$                                   |
| Geometric            | $(1-p)^x p,$<br>$x = 0, 1, \dots, 0 \leq p \leq 1$                                                                         | $\frac{p}{1-(1-p)e^t}, \quad t < \log(1/(1-p))$                                                           | $\frac{1-p}{p}$                             | $\frac{(1-p)}{p^2}$                                    |
| Poisson              | $\frac{e^{-\lambda} \lambda^x}{x!},$<br>$x = 0, 1, \dots, \lambda > 0$                                                     | $e^{\lambda(e^t-1)}$                                                                                      | $\lambda$                                   | $\lambda$                                              |
| Hypergeometric       | $\frac{\binom{A}{x} \binom{B}{n-x}}{\binom{A+B}{n}},$<br>$x = 0, 1, \dots, n$                                              | —                                                                                                         | $\frac{nA}{A+B}$                            | $\frac{nAB(A+B-n)}{(A+B)^2(A+B+1)}$                    |

Table 1: Summary of some common distributions in probability and statistics.